



Launch vehicles



Nose holds Soyuz or Progress spacecraft

Soyuz FG Russia 2001

Four booster rockets 64-ft (19.6-m-) tall fire at launch



Atlas V USA 2002

Fairing covers payload during launch, but opens to release craft or satellite once in orbit

Powerful boosters fall away four minutes after launch



Delta IV Heavy USA 2004

Spacecraft's emergency crew escape system

A Delta IV Heavy weighs more than **200 female elephants!**

Each rocket booster weighs 305 tons, when full of fuel



Ariane 5 Multinational 2005



Virgin Galactic SpaceShipTwo USA 2010

Rocket boosters fire for under 90 seconds at launch



Dream Chaser USA under development

Upturned wing for gliding back down to Earth

rocket engines, and falls away after its fuel is exhausted, leaving the remaining smaller, lighter vehicle to continue. The biggest lifter among current launch vehicles is the **Delta IV Heavy**, which can carry 31-ton loads into Earth orbit. This is just a quarter of the load carried by the

three-stage **Saturn V**, used for the Apollo Moon landings. Space planes, such as the **Space Shuttle Discovery** and the **SpaceShipTwo**, are powered by rocket engines but use their wings to glide back to the Earth after their mission.